trough emerging from the other, they cancel each other out. (In fact, this is the idea behind fancy noise-eliminating headphones—they measure the shape of the incoming sound wave and then produce another whose shape is exactly "opposite," leading to a cancellation of the undesired noise.) In between these extreme overlaps—peaks with peaks, troughs with troughs, and peaks with troughs—are a host of partial height augmentations and cancellations. If you and a slew of companions form a line of little boats parallel to the barrier and you each declare how severely you are jostled by the resulting water wave as it passes, the result will look something like that shown on the far right of Figure 4.7. Locations of significant jostling are where wave peaks (or troughs) from each slit coincide. Regions of minimal or no jostling are where peaks from one slit coincide with troughs from the other, resulting in a cancellation.

Since the photographic plate records how much it is "Jostled" by the incoming light, exactly the same reasoning applied to the wave picture of a light beam tells us that when both slits are open the photograph will look like that in Figure 4.8. The brightest areas in Figure 4.8 are where light-wave peaks (or troughs) from each slit coincide. Dark areas are where wave peaks from one slit coincide with wave troughs from the other, resulting in a cancellation. The sequence of light and dark bands is known as an interference pattern. This photograph is significantly different from that shown in Figure 4.6, and hence there is a concrete experiment to distinguish between the particle and the wave pictures of light. Young carried out a version of this experiment and his results matched Figure 4.8, thereby confirming the wave picture. Newton's corpuscular view was defeated (although it took quite some time before physicists accepted this). The prevailing wave view of light was subsequently put on a mathematically firm foundation by Maxwell.

But Einstein, the man who brought down Newton's revered theory of gravity, seems now to have resurrected Newton's particle model of light by his introduction of photons. Of course, we still face the same question: How can a particle perspective account for the interference pattern shown in Figure 4.8? At first blush you might make the following suggestion. Water is composed of H2O molecules—the "particles" of water. Nevertheless, when a lot of these molecules